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Linear and non-linear optical properties of capped CdTe nanocrystals prepared by mechanical alloying

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Abstract

CdTe nanocrystals were prepared by mechanical alloying the elemental Cd and Te powders. The formation of CdTe with a single cubic phase after 20 h of ball milling was confirmed by X-ray diffraction (XRD). The surface of as-milled CdTe nanoparticles was then capped with polarization TOP/TOPO or $(Na_3PO_4)_n$ organic ligand, which resulted in colorful dispersion solution with optical absorption peaks located at 573 nm and 525 nm, respectively. The third-order non-linearity, namely, the non-linear refraction and two-photon absorption (TPA) coefficient, of the capped CdTe dispersion samples were evaluated using Z-scan technique. The fitting of Z-scan experimental data with a special equation demonstrated that the capped CdTe nanocrystals possess large third-order susceptibilities at resonant wavelength. The non-linear figure of merit (γ/β) for 20 h as-milled CdTe nanocrystals after capping with TOP/TOPO was determined to be $\sim -2 \times 10^{-5}$ m, which is nearly 215 times larger than the value reported for bulk CdTe crystals.
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1. Introduction

Quantum dots provide an opportunity to investigate the behavior of semiconductors such as CdTe in the finite size regime [1–4]. Due to the quantum confinement these materials have many fundamentally interesting and potentially useful optical properties [5–9]. Highly luminescent semiconductor nanocrystals (NC) have received increasing attention for use in light-emitting devices and tagging application [10–13]. In the early 90s, high quality CdTe nanocrystals have become available by using $Cd(CH_3)_2$ as the cadmium precursor and the technical grade trioctylphosphine oxide (Tech TOPO) as the reaction solvent [14–16]. The synthesis of CdSe and CdTe nanocrystals through the organometallic route has resulted in high quality semiconductor NC for more than 10 years [17–21]. However, some key chemicals used in this traditional route are extremely toxic, pyrophoric, explosive and expensive [20,21]. In recent years other wet chemical methods have been employed for the synthesis of CdTe and CdSe NC's using safe, low cost, inorganic compounds of CdO, $Cd(AC)_2$, and $Cd(CO_3)$ as cadmium precursors [22,5]. Thiol as the stabilizing agents was also employed for the aqueous synthesis of CdTe NC [16–19]. Recently, a dry physical route, namely mechanical alloying process, has been used to synthesize surface clean CdTe nanocrystals in larger scale [23]. Subsequent capping the surface of the particles with TOP/TOPO resulted in colorful dispersion solution with similar optical properties to those CdTe nanocrystals prepared by wet chemical method [23,5,19].

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Concurrently, the design and characterization of new non-linear optical materials have attracted a great deal of attention with the goal to develop materials possessing large non-linearities and satisfying various technological requirements for photonic device application [24–26]. Several experimental and theoretical studies have shown that semiconductor NC, such as GaAs [27], CdS_xSe_{1-x} [28], PbSe [29], and SnO_2 [30], possess large third-order optical non-linear susceptibilities and short response times, which suggest their potential application in non-linear photonic devices [31,32]. Glass is a suitable host matrix for the nanocrystals in optical applications including optical switches and memories [28,33,34]. Although investigations of the optical non-linearity of CdTe NC doped in glass have been reported by degenerate four-wave mixing techniques in the resonance region ($\hbar\omega > E_g$, E_g is the band gap of material) [35,36], the sign of the non-linear refractive index and non-linear absorption coefficient of the CdTe NC was not determined.

In this paper, the physically as-milled CdTe nanoparticles were capped on the surface by polarization organic ligands of TOP/TOPO, which were then dispersed in hexane. The linear optical properties (i.e. UV to visible transmission) of colorful dispersion solutions of capped CdTe nanocrystals is presented, together with their non-linear absorption and non-linear refractive index coefficients determined by Z-scan technique with open and closed aperture. Greatly improved third-order non-linear optical properties were observed for these capped CdTe nanoparticles, which were fabricated by mechanical alloying process before capping. The advantage of the physical process for synthesizing CdTe NC is that one can obtain light emitting materials with tunable emission from the ultraviolet (uncapped CdTe NC's) to visible wavelength range (capped CdTe nanocrystals) [23]. Also, the mechanical alloying method provides an economic and effective routine to produce CdTe NC in relatively large quantities.

2. Experimental

High purity cadmium (99.99%) and tellurium (99.999%) powders were mixed and sealed in a hardened steel vial with hardened steel balls. Milling balls with different diameters (2–12 mm) were used. The ball milling of Cd and Te powders was performed on a SPEX 8000D Mix/miller using a ball to powder mass ratio of 20:1. The stoichiometric amounts of Cd and Te elemental powder with 1:1 atomic ratio were then mechanical alloyed to CdTe compound after ball milling for a certain period. Small amounts of as-milled powders were taken out from the vial within different time intervals for structural and optical measurements. The structural phases of the as-milled powders were measured by a Rigaku powder X-ray diffractometer. The microstructure and morphology of capped and uncapped CdTe nanoparticles were observed using a JEOL 2010 transmission electron microscopy operating at 200 kV.

The as-milled CdTe NC for different time periods were subsequently capped on the surface by polarization organic ligands TOP/TOPO or sodium polyphosphate, which were then dispersed in mixture of hexane. Linear UV visible optical absorption measurements were carried out upon the dispersion solution of these capped CdTe nanocrystals.

Afterwards, the colorful dispersion solutions of the capped CdTe nanoparticles were used for third-order optical non-linearity measurements by Z-scan technique. The colorful solution containing capped CdTe nanoparticles was deposited onto the surface of a micro-quartz glass slide by a Spin Coater. The samples were coated on the surface of the glass slide in such a way that the surface of the slide is fully covered and no air bubbles were trapped between the solution and substrate interface. The coated glass slides were then put in a small vacuum chamber for drying.

The third-order non-linear susceptibilities of the coated films containing CdTe NC on the surface of glass slides were measured using a single beam Z-scan technique [25,36]. Using a single Gaussian laser beam in a focused geometry, the transmittances of a non-linear medium as a function of the sample position z with respect to the focal plane are related to the non-linear absorption and refraction of sample for the open and closed aperture geometries, respectively. Therefore the non-linear absorption and non-linear refractive index coefficients can be obtained from the Z-scan measurements. The excitation source was a ~1 ps Ti: Sapphire laser operating at a wavelength of ~800 nm, a repetition rate of 82 MHz with a good Gaussian intensity distribution. The laser beam was focused to a waist radius ~57 μm with a lens of focal length 75 mm. A typical peak power density was ~85 kW/cm^2. The thickness of the sample was ~90 μm, which satisfies the condition of an optically thin sample ($z_R = \pi\omega_0/\lambda > L$). The sample at the vicinity of beam waist was moved to the negative and positive directions of the propagation of laser light. The laser power transmission through the sample was measured with open and closed apertures in the far-field region.

3. Results and discussion

3.1. Structural characterization of as-milled CdTe nanocrystals

3.1.1. X-ray diffraction results

Cadmium and tellurium powders were mixed together in stainless steel vials and mounted on a SPEX

8000D ball-milling machine. CdTe nanoparticles were synthesized via the self-propagation reaction of Cd + Te → CdTe under high frequency impact of hardened alloy balls on the wall. The crystal structure of the mechanically alloyed products within different period was identified using an X-ray powder diffracto-meter.

Fig. 1 shows X-ray diffraction patterns for the ball-milling products of elemental Cd and Te powders for 4.5 h and 20 h, respectively. The detail structural development of the ball milling the elemental mixture powders was described elsewhere [23]. From Fig. 1(a) it can be seen that with increasing ball-milling time up to 4.5 h, the diffraction peaks from zinc blende CdTe became dominant, while the diffraction intensity indexed from elemental Cd and Te phases became subtle. After about 20 h of ball milling, all XRD peaks from elemental Cd and Te disappeared (Fig. 1(b)) and only three main diffraction peaks corresponding to the {1 1 1}, {2 0 0} and {3 1 1} lattice planes from cubic CdTe structure were observed. Therefore, the XRD results confirmed that single phase CdTe nanocrystals with zinc blende structure was successfully fabricated after mechanical alloying elemental powders of Cd and Te for 20 h. This result was also consistent with conclusions drawn from HRTEM observation [23]. Further ball-milling for longer period up to 50 h resulted only in the broadening of the full width at half maximum (FWHM) of the XRD peaks for CdTe nanopowders [23]. The broadening effect of the peaks was mainly due to finer particle size. According the Scherer equation, the average particle size of the as-milled CdTe samples for 4.5 h and 20 h was estimated to be 23 nm and 13 nm, respectively.

3.1.2. TEM images for capped CdTe nanoparticles

The morphology of TOP/TOPO capped CdTe nanoparticles dispersed in hexane solution is shown in Fig. 2.

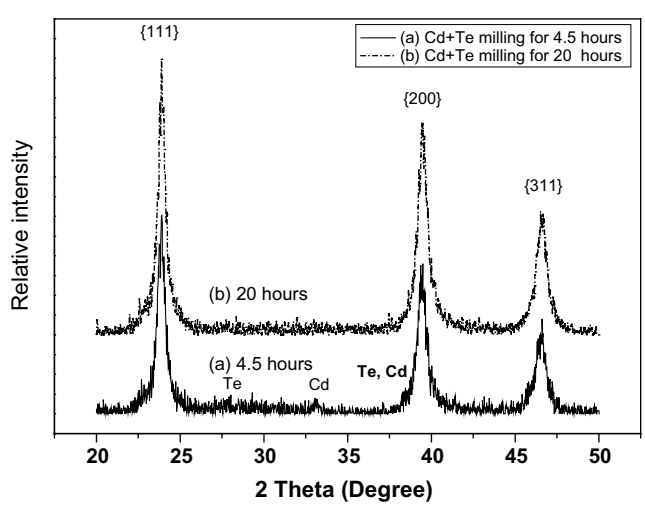

Fig. 1. XRD pattern for the mechanochemical reaction of elemental Cd and Te powder as a function of ball-milling time.

The corresponding bare CdTe nanoparticles before capping were prepared by ball milling the elemental Cd and Te mixture powders for 20 h. The average particle size of these capped CdTe nanocrystals was estimated to be within the range of 2–10 nm, which agrees roughly with the value derived from the UV visible absorption spectrum [23]. In contrast, without being capped by polarization TOP/TOPO organic agent, the as-milled powder was black, and its dispersion solution in hexane exhibited no color. The size distribution of the as-milled CdTe powders before capping inhomogeneously covered a wide range, over 2–50 nm, the particles also aggregated together and were difficult to be seen individually. After capping the surface of CdTe NC with polarization organic solvent TOP/TOPO, negative charges could be formed, which lead to a repulsive force among the capped nanoparticles. This repulsive force is strong enough to support the very small particles ($d < 10$ nm) floating in the solution, but not strong enough to support floating of larger particles (more than 10 nm). Therefore, large particles deposited onto the bottom of the vial regardless of whether being capped or not. The remaining very small capped CdTe nanoparticles within size range of 2–10 nm floated in the dispersion solution and then exhibited a homogeneous size distribution as being shown in the TEM image of Fig. 2.

3.2. Linear optical properties of capped CdTe nanoparticles

The uncapped, as-milled CdTe nanocrystals exhibited quite different optical properties compared to the capped CdTe NC, prepared by either physical method or chemical process [23]. The difference lies in the absorption peak wavelengths of the samples, with different colors of the dispersing solutions. For uncapped CdTe nanoparticles being dispersed in hexane or methanol solution, the dispersive solution was colorless. After being capped with polarization organic agent TOP/TOPO or sodium polyphosphate, the CdTe nanoparticles were then dispersed in non-polarization

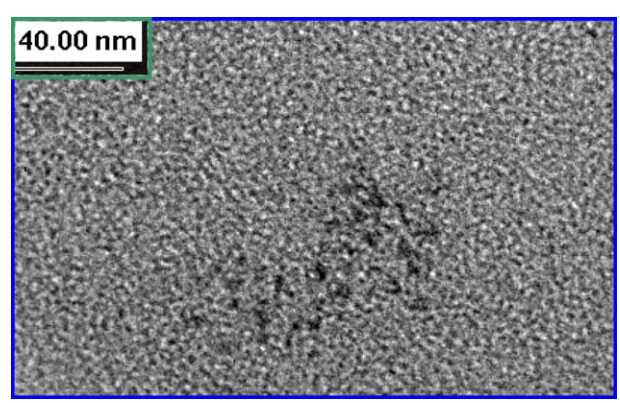

Fig. 2. TEM images of capped CdTe nanoparticles milled for 20 h.

organic hexane solvent, which made the colloid dispersive solution colorful. The color of the solution ranged from red, blue, to green and yellow depending on particle size. The colorful capped samples exhibited absorption peaks within the visible wavelength range, while the colorless uncapped samples had absorption peaks in the ultraviolet wavelength range [23]. The UV visible optical spectrum for the dispersion solution of TOP/TOPO capped CdTe nanoparticles is shown in Fig. 3(a), which exhibited a green color. While Fig. 3(b) is corresponding to the UV visible optical spectrum for the dispersion solution of sodium polyphosphate capped CdTe nanoparticles in water, which exhibited a red color. Both samples of bare CdTe powders before capping were fabricated by mechanical alloying the corresponding Cd and Te elemental powders for 20 h. From Fig. 3 it was observed that the UV visible absorption spectrum for CdTe-TOP/TOPO system showed up two absorption peaks at 530 nm and 573 nm, respectively. The CdTe–Na_3PO_4 system exhibited only one peak at around 525 nm. The difference between the two systems is due to the surface charge contribution arising from the different dielectric function of two polarization capping solvents, as being revealed by the S factor in Brus' formula [9]. The band gap energy of the capped CdTe NC derived from the absorption peak (2.34 eV, 2.16 eV and 2.36 eV) is much larger than the band gap energy of bulk CdTe (~1.48 eV), indicating that the particle size is extremely small, i.e. within 2–8 nm, which agrees with TEM evaluation results. The smooth sharp absorption peaks in Fig. 3 indicates that the size distribution of the CdTe NC is homogeneous after being capped by either TOP/TOPO or sodium polyphosphate, comparable to that being observed in those samples prepared by chemical methods [15].

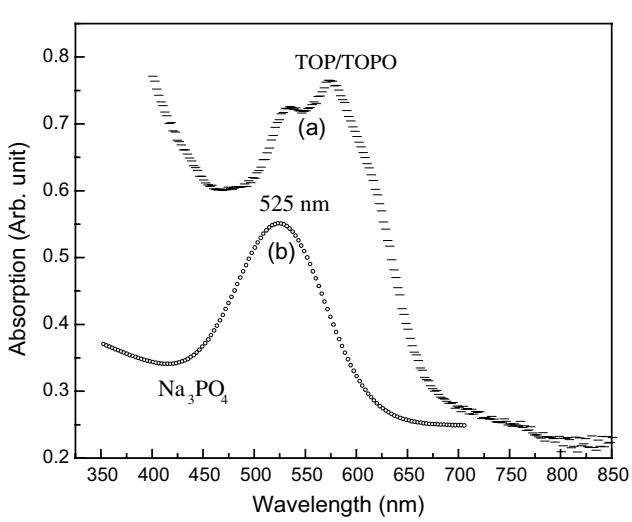

Fig. 3. UV visible absorption spectrum for CdTe nanoparticles ball milling for 20 h, (a) capped by TOP/TOPO in hexane and (b) capped by Na_3PO_4 in water.

3.3. Non-linear optical properties

3.3.1. Z-scan with open aperture

The laser power transmission through the sample was measured with open and closed apertures in the far-field region. The sample preparation and non-linear optical measurement process were described in Section 2. The normalized transmittance for the capped CdTe sample ball milled for 20 h as a function of the distance from the beam waist is shown in Fig. 4 without collecting aperture.

The Z-scan technique is a simple but sensitive method by which the optical non-linearity of the materials can be measured [37]. Important information can be derived form the Z-scan. First, since the peak of the normalized transmittance for the closed aperture Z-scan procedure hump like valleys, the sign of the refractive index is negative, indicating a self-defocusing material. Secondly, the shape of the open aperture Z-scan in Fig. 4 strongly suggests that non-linear absorption occurred in this system. The non-linear absorption coefficient is approximately given by the following equation [38]:

$$T_{\text{open}} = (1-R)^2 e^{-\alpha L} \sum_{m=0}^{\infty} \frac{[-(1-R)\beta I_0 L_{\text{eff}}]^m}{(1+m)^{1.5}} \quad (1)$$

where R is the reflectivity, α is the linear loss, L is the sample thickness, I_0 is the on-axis incident laser beam peak intensity, $L_{\text{eff}} = (1 - e^{-\alpha L})/\alpha$ is defined as effective thickness of the sample, where the condition $|(1-R)\beta I_0 L_{\text{eff}}| < 1$ should be satisfied.

The consistent fitting to the experimental data using Eq. (1) gives the non-linear absorption coefficient β.

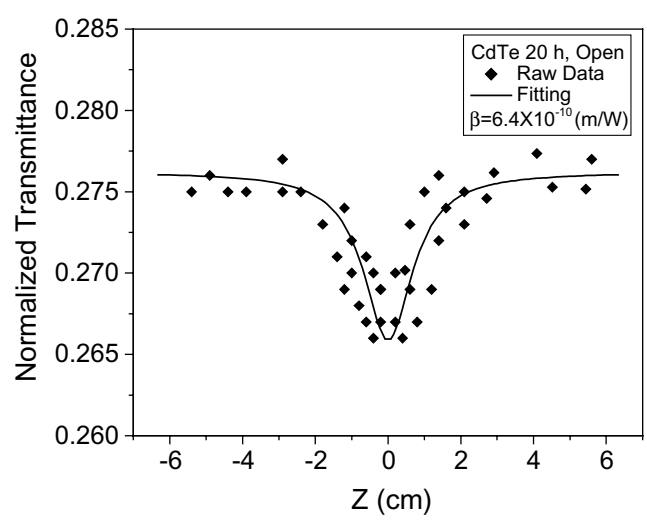

Fig. 4. Open Z-scan results for capped CdTe nanoparticles dispersed in hexane. The solid line is the best fitting of Eq. (1) to the raw data. The nonlinear absorption obtained from Z-scan measurement is 6.4×10^{-10} m/W.

Using the parameters given above, the non-linear absorption coefficient after fitting to Eq. (1) yielded a value of $\beta = \sim 6.4 \times 10^{-10}$ m/W for 20 h capped CdTe nanocrystals. The positive sign of β indicates the non-linear absorption. The fluctuation of the data is caused by the unstable output of the laser. Due to the fact that the excitation CdTe nanocrystals by the laser satisfies the condition of $1/2E_g(=0.85\,\text{eV}) < \hbar\omega(=1.55\,\text{eV}) < E_g(=1.7\,\text{eV})$ [30], then two photon absorption (TPA) can take place, assuming that the band gap of CdTe nanoparticles is blue shifting to 1.7 eV from its bulk value of 1.48 eV when the average particle size is approaching to less than 5 nm under quantum confinement effect. The non-linear absorptions in all the capped CdTe samples with particle size less than 5 nm are therefore TPA process.

3.3.2. Z-scan with closed aperture

Once β is known from the open aperture Z-scan measurement, the non-linear refractive index coefficient γ can be extracted from the best fitting the closed Z-scan measurement results with a special equation, which is given by [39,40]:

$$T_{\text{on-axis}}(I_0,x) = 1 + \frac{I_0 L_{\text{eff}}}{\sqrt{2}}\frac{4kx\gamma - (3+x^2)\beta}{(1+x^2)(9+x^2)} + \frac{I_0^2 L_{\text{eff}}^2}{\sqrt{3}}\left[\frac{k^2\gamma^2 + \frac{\beta^2}{4}}{(1+x^2)(9+x^2)} - \frac{(5+x^2)\left(k^2\gamma^2 - \frac{3}{4}\beta^2\right) + 8kx\gamma\beta}{(1+x^2)^2(25+x^2)}\right] + \frac{I_0^3 L_{\text{eff}}^3}{\sqrt{4}}\left[\frac{1}{3}\frac{(7+x^2)\left(\frac{9}{2}k^2\gamma^2 - \frac{15}{8}\beta^2\right)\beta - 6k\gamma\left(k^2\gamma^2 - \frac{23}{4}\beta^2\right)x}{(1+x^2)^3(49+x^2)} + \frac{\left[\frac{1}{2}(15+x^2)\beta + 2k\gamma x\right]\left(k^2\gamma^2 - \frac{3}{4}\beta^2\right) + 2k\gamma\beta^2 x - 2(15+x^2)k^2\gamma^2\beta}{(1+x^2)^2(9+x^2)(25+x^2)}\right] \quad (2)$$

where $x = z/z_0$ is the relative distance from focus, k is the magnitude of the wave vector of the incident laser light, I_0 the peak intensity of the incident laser light beam.

We made certain measurements on the third-order non-linear susceptibilities for these capped CdTe nanoparticles suspended in hexane solvent, using a 1 ps Ti: Sapphire laser source with wavelength at 800 nm, at frequency of 82 MHz. The linear optical absorption spectrum in Fig. 3 does not show any absorption at 800 nm for the laser wavelength. Very good Z-scan results were obtained for these homogenous materials.

Figs. 5 and 6 show two typical closed Z-scan results at 800 nm for capped CdTe nanocrystals, which were prepared by ball-milling of elemental Cd and Te mixture powders for 20 h and 50 h respectively, subsequently being capped on the surface with TOP/TOPO, and then dispersed in hexane. The non-linear refractive index coefficient γ was obtained from best fitting the closed Z-scan results by Eq. (2), which reaches as high as -1.2×10^{-15} m^2/W for 20 h sample and -1.23×10^{-15} m^2/W for 50 h sample respectively. Taking into account that the best fitting to the open Z-scan results in Fig. 4 using Eq. (1) gives the result of $\beta = 6.4 \times 10^{-10}$ m/W, the non-linear figure of merit (γ/β) for the 20 h capped CdTe nanocrystals is therefore evaluated to be $\sim -2 \times 10^{-5}$ m (or -2×10^{-3} cm), which is approximately 10 times higher than that derived from Yu's measurement for CdTe nanoparticles doped glass

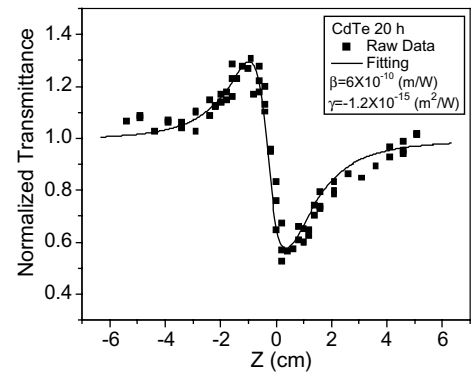

Fig. 5. Closed Z-scan results of capped CdTe nanoparticles ball milling for 20 h suspended in hexane. The solid line is the result of the best fitting of Eq. (2) to the experimental data. The γ parameter of -1.2×10^{-15} m^2/W is extracted from the fitting process.

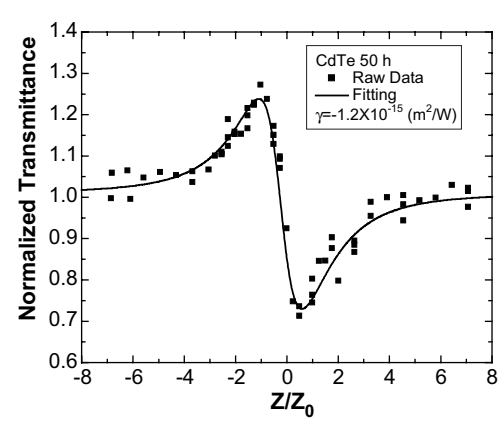

Fig. 6. Normalized transmittance of capped CdTe nanocrystals using a Z-scan with collecting aperture. The sample was ball milling for 50 h, then capped by TOPO/TOP surface agent and dispersed in Hexane solvent.

with 3–10 nm particle size range [36]. The γ and β values for bulk CdTe single crystal obtained by similar measurement method are $-2.4 \times 10^{-17}\,m^2/W$ and 2.6×10^{-8} cm/W respectively [25], the merit of figure for bulk CdTe is therefore -9.3×10^{-6} cm. Thereby the third-order non-linear merit of figure (γ/β) for 20 h as-milled CdTe nanocrystals being capped by TOP/TOPO is enhanced by a factor of 215 times compared to that for bulk CdTe crystals.

4. Conclusion

CdTe nanocrystals were fabricated by mechanical alloying the elemental Cd and Te powders for more than 20 h. The as-milled CdTe nanoparticles were then capped on the surface by polarization TOP/TOPO or sodium polyphosphate ligands. TEM images demonstrated homogeneous size distribution for these capped CdTe nanoparticles in hexane solution. The colorful dispersion solutions containing 20 h capped CdTe nanocrystals exhibited absorption peaks at 573 nm and 530 nm for TOP/TOPO capping ligand and 525 nm for sodium polyphosphate capping agent, respectively. Both the non-linear absorption and non-linear refractive index of the capped CdTe nanocrystals were investigated at 800 nm by using a 1 ps pulse width Ti: Sapphire laser with a repetition rate of 82 MHz. The results revealed that the non-linear refractive index of $\gamma = -1.2 \times 10^{-15}\,m^2/W$, the non-linear absorption coefficient, was determined to be $\beta = \sim 6.4 \times 10^{-10}$ m/W for 20 h capped CdTe nanocrystals, and its non-linear figure of merit (γ/β) was therefore evaluated to be $\sim -2 \times 10^{-5}$ m (or -2×10^{-3} cm), which is 215 times of that for bulk CdTe crystals. The large third-order non-linear susceptibility and great non-linear figure of merit indicate that as-milled CdTe nanocrystals capped by TOP/TOPO are excellent candidate material for photonic device applications.

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